

Making decisions

Programs make decisions about what to do by comparing variables, numbers, and strings using Boolean expressions. These give an answer of either “True” or “False”.

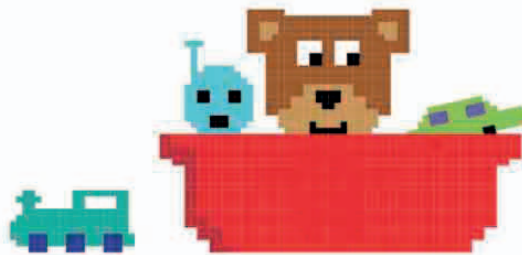
SEE ALSO

◀ **62–63** True or false?

◀ **108–109** Variables in Python

Logical operators

Logical operators are used to compare variables against numbers or strings, or even against other variables. The resulting answer is either “True” or “False”.



==

“Equals” operator

!=

“Not equal to” operator

<

“Less than” operator

>

“Greater than” operator

<=

“Less than or equal to” operator

>=

“Greater than or equal to” operator

△ Types of comparison operators

There are six comparison operators. Python uses two equals signs to compare if two things are the same. (A single equals sign is used to assign a value to a variable.)

▷ Use the shell to check

Logical operators also work in the shell window. Use this example to try out several logical operators, including “not”, “or”, and “and”.

```
>>> toys = 10
>>> toys == 1
False
>>> toys > 1
True
>>> toys < 1
False
>>> toys != 1
True
>>> toys <= 10
True
>>> not toys == 1
True
>>> toys == 9 or toys == 10
True
>>> toys == 9 and toys == 10
False
```

This checks whether “toys” is equal to 1

This checks whether “toys” is more than 1

This checks whether “toys” is less than 1

This checks if “toys” is not equal to 1

This checks if “toys” is less than or equal to 10

The “not” logical operator reverses the answer (in this example, from “False” to “True”)

The “or” logical operator checks if “toys” is 9 or 10

The “and” logical operator is used to check if “toys” is both 9 and 10. This can never be true, so the answer is “False”